Assignment 3

Apply ID3 algorithm to obtain a decision tree for the following data

Instance	Attribute-1	Attribute-2	Class
1	T	T	+
2	T	T	+
3	T	F	-
4	F	F	+
5	F	T	-
6	F	T	-

⇒ Total + = 3

Total - = 3

$$\text{Entropy} = -\left[P_+ \log_2 P_+ + P_- \log_2 P_-\right]$$

$$= -\left[\frac{3}{6}\log_2\frac{3}{6} + \frac{3}{6}\log_2\frac{3}{6}\right]$$

$$= 1$$

Attribute_1

$$\text{Entropy}(A\_1 = T) = -\left[\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right]$$

$$= 0.918$$

$$\text{Entropy}(A\_1 = F) = -\left[\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right]$$

$$= 0.918$$

$$Gain(At\_1) = E(S) - \sum_{v \in A} \frac{|S_v|}{|S|} E(S_v)$$

$$= 1 - \left(\frac{3}{6} \times 0.918 + \frac{3}{6} \times 0.918\right)$$

$$= 0.082$$

Attribute_2

$$Entropy(A\_2 = T) = -\left(\frac{2}{4}\log_2 \frac{2}{4} + \frac{2}{4}\log_2 \frac{2}{4}\right)$$

$$= 1$$

$$Entropy(A\_2 = F) = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{2}\log_2 \frac{1}{2}\right)$$

$$= 1$$

$$Gain(At\_2) = 1 - \left(\frac{4}{6} \times 1 + \frac{2}{6} \times 1\right)$$

$$= 1 - 1 = 0$$

$$Gain(At\_1) > Gain(At\_2)$$

The next split attribute is attribute_1

Attribute_1 = T

Attribute_2	class
T	+
T	+
F	-

$$Entropy(At\_1, At\_2 = T) = -\left(\frac{2}{2}\log_2 \frac{2}{2} + 0\right) = 0$$

$$Entropy(At\_1, At\_2 = F) = -\left(\frac{1}{1}\log_2 \frac{1}{1}\right) = 0$$

Attribute_1 = F

Attribute_2	class
F	+
T	-
T	-

$$\text{Entrophy}(At\_1, At\_2 = T) = -\left(\frac{1}{1}\log_2\frac{1}{1} + 0\right) = 0$$

$$\text{Entropy}(At\_1, At\_2 = F) = -\left(0 + \frac{2}{2}\log_2\frac{2}{2}\right) = 0$$

Since both branches become pure after splitting on Attribute_2, the tree terminates here (Entropy = 0 for every leaf).

Decision tree